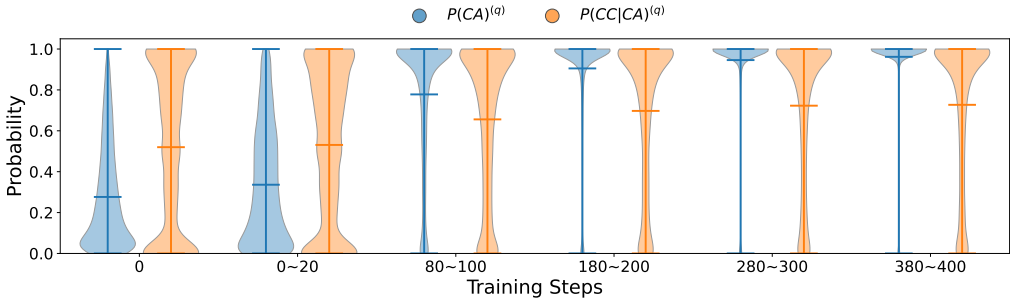
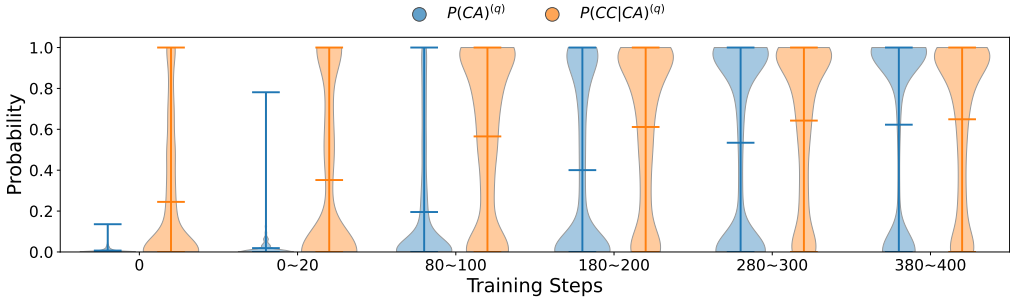
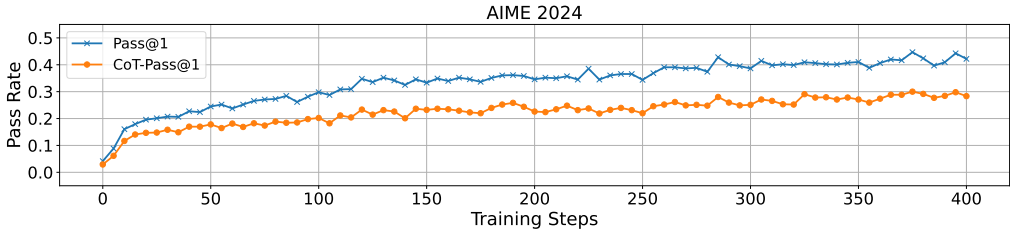


(a) Distributions of $P(CA)^{(q)}$ and $P(CC|CA)^{(q)}$ for **easy** training questions in DAPO.



(b) Distributions of $P(CA)^{(q)}$ and $P(CC|CA)^{(q)}$ for **hard** training questions in DAPO.



(c) Generalization performance on AIME 2024 across different training steps.

Figure 10: We show gradually optimized performance on training prompts throughout RLVR and corresponding generalization behaviors on testing prompts. The top two subfigures use violin plots to visualize the evolution of $P(CA)^{(q)}$ (the fraction of correct answers for prompt q) and $P(CC|CA)^{(q)}$ (the fraction of correct CoTs within the correct answers for prompt q) over the course of DAPO training. Subfigure (a) shows results for "easy" training questions, while (b) presents the corresponding distributions for "hard" questions ($P(CA)^{(q)}$ ¡ 1 for DAPO). We divide training questions into easy and hard by sampling 64 rollouts for each of the $17k$ training questions using Qwen2.5-32B, and labeling those with at least one correct answer as easy. Subfigure (c) presents the generalization performance on AIME 2024 across different training steps.

Call for Lightweight yet Powerful CoT Verifiers While DeepSeek-R1-0528-Qwen3-8B serves as a useful CoT verifier, it is not infallible. Conflicting verification results across multiple queries reveal the challenges of false-positive and false-negative verifications. To tackle this, we combine multiple verification strategies, including different voting rules, to improve robustness. Looking forward, there is a pressing need for light yet reliable CoT verifiers that can serve as standardized evaluators beyond the coarse-grained Pass@K metric. This direction also relates to previous studies on process reward modeling (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024).

Scaling RLVR or Scaling Pre-Training While the scaling of pre-training has led to transformative progress in LLMs (Kaplan et al., 2020; Liu et al., 2024), enabling the transition to the era of artificial general intelligence, we argue that scaling RLVR could be equally pivotal, given the empirical evidences and theoretical foundation that all demonstrate its real incentivization beyond base LLMs. As modern LLMs approach the limits of language token exposure, learning from experi-

ence (Silver & Sutton, 2025) may represent the next leap. Recent efforts by leading research teams suggest a growing emphasis on this direction (Guo et al., 2025; DeepSeek, 2025; Gemini, 2024; Grok, 2025; OpenAI, 2025; Qwen, 2025; Gemini, 2025; Anthropic, 2025; Mistral.AI, 2025). For the broad open research community, understanding the foundations and limitations of current RLVR algorithms is crucial to push this direction further.

New RLVR Algorithms and Beyond With our insight that RLVR implicitly incentivizes correct reasoning in base LLMs, we anticipate the development of new algorithmic paradigms. These may include optimization formulations or objective functions, such as policy-gradient approaches (Sutton et al., 1999; Schulman et al., 2017), new likelihood-based optimization objectives (Chen et al., 2025a; Zhu et al., 2025), and preference optimization frameworks (Rafailov et al., 2023; Su et al., 2025). The key principle is that the new algorithms should be designed to more directly incentivize correct reasoning paths, alleviating inherent logical biases in base LLMs.

A.8 CASE STUDIES

We perform manual verification on difficult problems from multiple benchmarks, focusing on cases where the base model occasionally produces a correct final answer despite flawed reasoning in its CoT. For each case study, we present the original question, the full response generated by Qwen2.5-32B, and a detailed breakdown of key errors identified by DeepSeek-R1-0528-Qwen3-8B, explicitly linking each error to the corresponding flawed reasoning segments in the CoT. For layout and color schemes, we follow the visualization strategies used in Yue et al. (2025).

A.8.1 CASE STUDIES ON AIME 2024

Question

AIME24 I Question No.9 (dataset id 74)[7]
Let A, B, C, and D be points on the hyperbola $\frac{x^2}{20} - \frac{y^2}{24} = 1$ such that $ABCD$ is a rhombus whose diagonals intersect at the origin. Find the greatest real number that is less than BD^2 for all such rhombi.

Qwen2.5-32B Answer

The diagonals of a rhombus are perpendicular and bisect each other, so $ABCD$ is a square. Let $a = BD = AC$.

> DeepSeek-R1-0528-Qwen3-8B verify: - **Conceptual Error**: The solution incorrectly assumes that ABCD is a square, whereas the problem only specifies that it is a rhombus. A rhombus does not necessarily have equal diagonals, so this assumption is invalid.

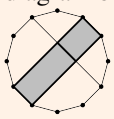
Then $a^2 = 2(AB^2)$. The equation of the hyperbola can be rewritten as $y^2 = 24\left(\frac{x^2}{20} - 1\right) = \frac{6}{5}x^2 - 24$. Let P be the point where the diagonals intersect. Then $OP = \frac{a}{2}$, so $P = \left(\frac{a}{2}, \frac{a}{2}\right)$.

> DeepSeek-R1-0528-Qwen3-8B verify: - **Logical Error**: The solution defines P as the intersection point and sets it to (a/2, a/2), but the problem states that the diagonals intersect at the origin. This inconsistency leads to incorrect reasoning, as P should be (0,0) based on the problem statement.

Since P lies on the hyperbola, we have $\frac{a^2}{4} = \frac{6}{5}\left(\frac{a^2}{4}\right) - 24$, so $\frac{a^2}{20} = 24$, or $a^2 = 480$. Therefore, the greatest real number that is less than BD^2 for all such rhombi is $\boxed{480}$.

Question

AIME24 II Question No.15(dataset id 81)[8]
Find the number of rectangles that can be formed inside a fixed regular dodecagon (12-gon) where each side of the rectangle lies on either a side or a diagonal of the dodecagon. The diagram below shows three of those rectangles.



[7] https://artofproblemsolving.com/wiki/index.php/2024_AIME_I_Problems/Problem_9

[8] https://artofproblemsolving.com/wiki/index.php/2024_AIME_II_Problems/Problem_15